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Investigating the Genetic Relationship Between Vitamin B12 Metabolism and Parkinson Disease

Raphael Dering,[1,*] Margarita Onvumere,[1,*] Lang Liu,[2,3,4] Philippe Huot,[2,5,6] Ziv Gan-Or,[2,3,4] and Konstantin Senkevich[2,3,4,7]

Correspondence
Dr. Senkevich
konstantin.senkevich@mcgill.ca

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Abstract

Background and Objectives

Epidemiologic studies suggest that patients with Parkinson disease (PD) may have lower levels of vitamin B12 compared with healthy controls, and it was proposed that patients with PD could benefit from vitamin B12 supplementation. Functional studies have shown that B12 could modify LRRK2 activity and may directly interact with alpha-synuclein. The aim of this study was to investigate the role of common and rare variants in genes related to B12 metabolism and assess the potential causal relationships between B12 levels and PD risk, age at onset, and motor/cognitive progression.

Methods

We investigated the association between common and rare variants in genes involved in vitamin B12 metabolism. Rare variants (minor allele frequency <0.01) were analyzed using the optimal sequence kernel association test in 4,815 patients with PD and 65,607 controls from 2 independent cohorts. We constructed pathway-specific polygenic risk scores (PRSs) for genes essential to B12 metabolism and for genes identified in previous genome-wide association studies (GWASs) on B12 metabolism. Mendelian randomization and genetic correlation analyses were applied to explore the relationship between vitamin B12 levels and PD risk, age at onset, and disease progression.

Results

Our analysis showed no associations between common variants of genes crucial in B12 metabolism and PD. Pathway PRSs identified nominal association between B12-related genes and PD (odds ratio [OR] = 1.061, 95% CI 1.004–1.121, p = 0.038), which did not survive Bonferroni correction. In the rare-variant analysis, we identified a significant association between variants with high Combined Annotation Dependent Depletion scores in the *CUBN* gene (p = 6.07E-05; Pfdr = 0.005) in the Accelerating Medicines Partnership-PD cohort, driven by the benign variant p.G3114S (OR = 3.3; p = 3.56E-05); however, this was not validated in the meta-analysis. We did not identify a potentially causal relationship between vitamin B12 levels and the risk, age at onset, or progression of PD. In addition, no genetic correlation was observed between vitamin B12 and PD risk or age-at-onset GWASs.

Discussion

Overall, our analyses indicate lack of genetic link between B12 levels or metabolism and PD.

Introduction

Multiple ongoing clinical trials are investigating the efficacy of dietary adjustments in Parkinson disease (PD).[1] Several epidemiologic studies have shown that vitamin B12 levels tend to be

*These authors contributed equally to this work.

[1]Institute of Applied Computer Science, ITMO University, Saint Petersburg, Russia; [2]The Neuro (Montreal Neurological Institute-Hospital), McGill University, Quebec, Canada; [3]Department of Neurology and Neurosurgery, McGill University, Montréal, QC, Canada; [4]Department of Human Genetics, McGill University, Montréal, QC, Canada; [5]Department of Neurology and Neurosurgery, McGill University, Montreal, QC, Canada; [6]Division of Neurology, Department of Neuroscience, McGill University Health Centre, Montreal, QC, Canada; and [7]Department of Specialized Medicine, Division of Medical Genetics, McGill University Health Centre, Montreal, QC, Canada.

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Glossary

AMP-PD = Accelerating Medicines Partnership-PD; **CADD** = Combined Annotation Dependent Depletion; **GWAS** = genome-wide association study; **LD** = linkage disequilibrium; **MAF** = minor allele frequency; **MR** = Mendelian randomization; **OR** = odds ratio; **PD** = Parkinson disease; **SKAT** = sequence kernel association test; **SNV** = single nucleotide polymorphism; **UKBB** = UK Biobank.

lower in patients with PD than in healthy individuals, suggesting that vitamin B12 supplementation could be beneficial in PD.[2-4] Functional studies have demonstrated that B12 could inhibit LRRK2 kinase activity and directly interact with alpha-synuclein.[5,6] However, no genetic studies have causally linked B12 level or metabolism and PD or identified genetic factors leading to potentially lower B12 in patients with PD.[7]

The human body relies entirely on external sources for vitamin B12, because it cannot synthesize the vitamin internally. However, there are several genes involved in its transport and metabolism (function detailed in Table 1). These genes regulate the absorption, distribution, and conversion of B12 into its active forms.[8,9] Given the emerging evidence linking low B12 levels to PD, it is important to understand whether genetic variations in B12 metabolism may contribute to the risk and progression of PD.

The aim of our study was to investigate the potential role of genes involved in B12 metabolism in PD and assess whether there is a potential causal relationship between B12 levels and the risk, age at onset, and progression of PD. First, we analyzed common and rare variants in genes related to B12 metabolism. We then constructed pathway-specific polygenic risk scores (PRSs) based on the genes essential to B12 metabolism and all nominated genes identified from the previous genome-wide association studies (GWASs) on B12 metabolism. In addition, we used Mendelian randomization (MR) to assess whether there is a causal relationship between B12 levels and the risk, age at onset, and motor and cognitive progression of PD, using GWASs of B12 level as a proxy to assess the genetic determinants of B12 levels and PD risk.

Methods

Study Population

To investigate whether common variants in genes involved in vitamin B12 metabolism are linked to PD, we analyzed summary statistics from the largest available PD GWAS of European ancestry.[10] Locus zoom plots were generated using the locuszoomr R library.[11] For pathway-specific PRS analysis, 7 independent cohorts were used, as detailed in eTable 1.

For the analysis of rare variants, we used whole-genome sequencing data from 2 independent cohorts, including 4,929 patients with PD and 68,029 controls. Whole-genome sequencing data were obtained from the Accelerating Medicines Partnership-PD (AMP-PD) initiative (detailed in the Acknowledgments), and from the UK Biobank (UKBB) through the Neurohub platform (eTable 2).

For MR, Single Nucleotide Variants (SNVs) derived from the GWAS on vitamin B12 serum levels were used as exposure.[12] The largest European PD risk GWAS (excluding 23andMe, Inc. data) and a PD age-at-onset GWAS were used as

Table 1 Genes Involved in B12 Metabolism

Gene	Encoded protein	Chr:position (hg38)	Function
AMN	Amnion associated transmembrane protein	chr14:102,922,663-102,930,842	Part of a complex for vitamin B12 absorption in the intestines
CUBN	Cubilin	chr10:16,823,966-17,129,811	Receptor that binds the B12-intrinsic factor complex for absorption
MMACHC	Methylmalonic aciduria and homocystinuria, cobalamin C type	chr1:45,500,300-45,513,382	Processes cobalamin into its active forms for cellular use
MTR	Methionine synthase	chr1:236,795,292-236,903,981	Catalyzes homocysteine to methionine conversion using methylcobalamin
MTRR	Methionine synthase reductase	chr5:7,869,148-7,901,113	Regenerates methylcobalamin for methionine synthase function
TCN2	Transcobalamin 2	chr22:30,607,174-30,627,271	Transports vitamin B12 in the bloodstream to tissues
FUT2	Fucosyltransferase 2	chr19:48,695,971-48,705,951	Involved in the biosynthesis of the H antigen, influencing the gut microbiota and indirectly affecting vitamin B12 absorption

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outcomes.[10,13] Controls in the PD risk GWAS were population-based, and no biochemical screening for factors that could alter vitamin B12 metabolism (e.g., malabsorption, anemia, and supplementation) was reported.

We investigated the potential causal association between vitamin B12 and PD progression, using summary statistics from a GWAS meta-analysis of 12 longitudinal cohorts (4,093 patients; median follow-up 3.81 years).[14] Continuous traits such as Mini-Mental State Examination scores, Part III Unified PD Rating Scale scores, and Hoehn and Yahr Scale scores were modeled as yearly slopes in linear mixed-effect analyses. The binomial outcome, incident cognitive impairment, was analyzed with baseline logistic regression and then with a Cox time-to-event model once at least 20 new cases occurred. The source GWAS adjusted for levodopa and dopamine-agonist exposure as time-varying covariates and z-standardized scores within cohorts before meta-analysis.[14] Follow-up schedules varied among cohorts, so no single fixed interval was imposed. Effective sample size for each progression GWAS was calculated as the mean values across all SNVs in the summary statistics. All individuals in the included GWAS were of European ancestry (eTable 3).

Gene Selection

For the core analysis of common and rare variants, we selected essential genes representing major steps of B12 metabolism (Figure 1; Table 1): intestinal uptake (*AMN*, *CUBN*), intracellular processing (*MMACHC*), cytosolic remethylation (*MTR*, *MTRR*), plasma transport (*TCN2*), and systemic regulation (*FUT2*).[9] For the PRS analysis (eTable 4), we expanded the list to include all loci significantly associated with B12 metabolism and extracted from the GWAS Catalog,[15] covering further canonical components of the pathway: gastric binding (*TCN1*), lysosomal export (*ABCD4*), mitochondrial metabolism (*MMAA, MMUT*), tissue uptake (CD320), and glycosylation (*FUT3*), as well as additional GWAS-implicated loci with less clearly defined roles. Together, these selections capture the major stages of B12 absorption, transport, and utilization.[9]

Pathway-Specific PRS

To investigate potential genetic association of genes involved in vitamin B12 metabolism with PD, we calculated 2 pathway-specific PRSs, one based on a curated set of core genes and another based on an extended list including all loci previously associated with B12 metabolism in GWASs.

We calculated PRSs in participants of European ancestry from 7 cohorts (eTable 1), excluding first-degree and second-degree relatives, as previously described.[16] In brief, we performed a sex discrepancy check by comparing recorded biological sex with genetically inferred sex. For PRS calculation, common SNVs with a minor allele frequency (MAF) > 0.01 and p value <0.05 were included, and we performed linkage disequilibrium (LD) clumping to exclude variants with r^2 > 0.1 within 250 kb. Covariates such as age at onset for patients with PD, age at enrollment for controls, sex, and the top 10 principal components were included in the regression models to control for confounding factors. To test for a possible interaction between vitamin B12 pathways and *LRRK2*, we repeated the regression with carrier status for the pathogenic variant p.G2019S added as an additional covariate. A permutation test with 10,000 repetitions was conducted to generate empirical p values for the genetic pathways of interest.

Rare-Variant Analysis

To explore the role of rare variants in genes involved in vitamin B12 metabolism (Table 1), we used genetic data from whole-genome sequencing data sets (UKBB and AMP-PD). For both cohorts, genetic data were aligned to the human reference genome build hg38. We included only participants of European ancestry and excluded first-degree and second-degree relatives. Quality control for the AMP-PD whole-genome sequencing data was performed following established protocols (detailed in).[17] In brief, we included samples with a minimum average coverage of ×25 and excluded those with more than 5% missing genotypes. For whole-genome sequencing data from the UKBB, we applied the Genome Analysis Toolkit (GATK, v3.8) for quality control. We included samples with a minimum depth of coverage of ×10 and a genotype quality score threshold of 20.

We conducted rare-variant association analysis (MAF <0.01), using the optimized sequence kernel association test (SKAT-O).[18] Variants were grouped into categories: all rare variants, nonsynonymous variants, loss-of-function variants (stop-gain, frameshift, or canonical splice-site), and variants with a Combined Annotation Dependent Depletion (CADD) score ≥20, predicted to be highly deleterious. Meta-analysis of the 2 cohorts was performed using the metaSKAT R package.[19]

MR and Genetic Correlation

For the construction of genetic instruments in MR analysis, we extracted SNVs with GWAS-level of associations (p value <5.0E-08) from the exposure GWAS on B12 level.[12] Selected SNVs were then clumped using standard parameters (clumping window of 10,000 kb, r^2 cutoff 0.001) to exclude variants in LD. To estimate the strength of genetic instruments, we calculated R^2, which reflects the proportion of variability explained by genetic variants, and F-statistics.[20] We then calculated power using the online MR power calculator to detect an equivalent effect, represented by an odds ratio (OR) of 1.2.[21]

To perform MR, we used the Two Sample MR R package.[22] We performed Steiger filtering to exclude SNVs that explain more variance in the outcome than in the exposure.[23] We then used the inverse variance–weighted method, which combines results from individual Wald ratios together.[24,25] We applied the MR Egger test to identify net directional pleiotropy and obtain a more accurate estimate of the true causal effect. Heterogeneity was assessed using the Cochran Q-test. In addition, we performed an MR-PRESSO test to identify

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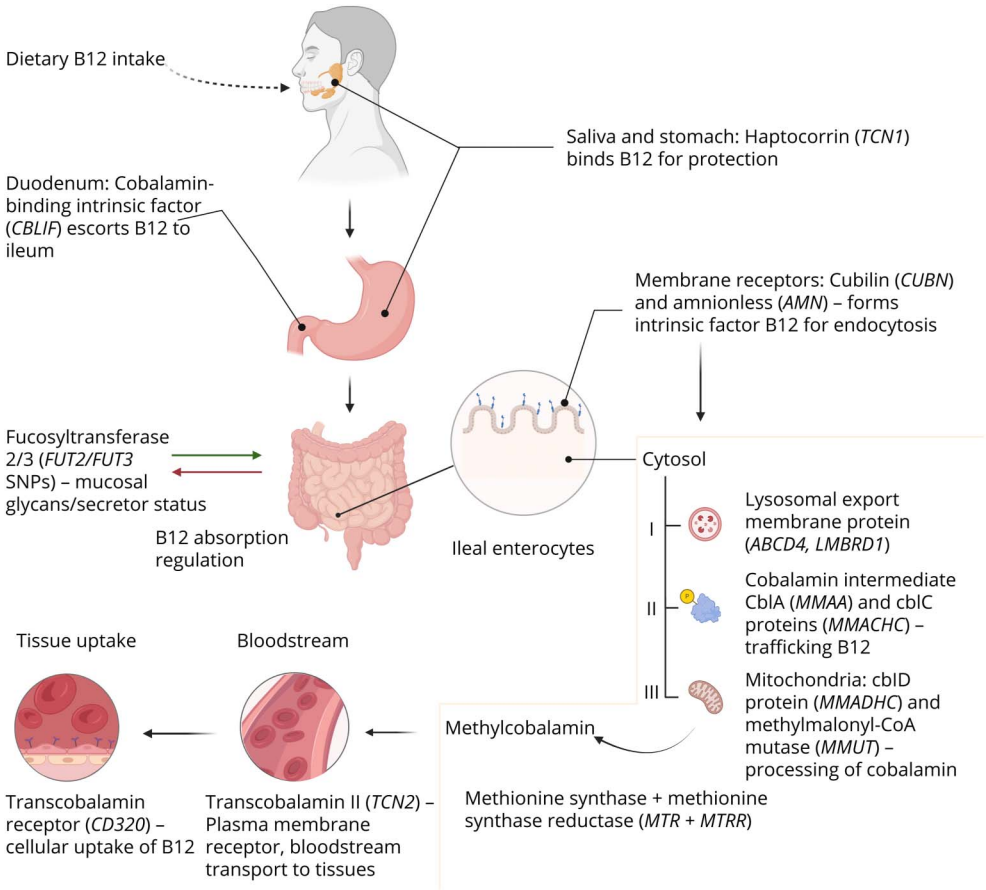
Figure 1 Vitamin B12 Absorption, Transport, and Intracellular Metabolism



Dietary B12 binds carrier proteins and is absorbed in the ileum via the cubilin-amnionless receptor, exported from lysosomes, and converted into active cofactors in the cytosol and mitochondria. Key genes are shown at each step. Created in BioRender. Dering, R. (2025) BioRender.com/74v0bgl.

pleiotropic outlier SNVs and adjust the pooled estimate if necessary.[26] We then performed genetic correlation analysis between PD risk and age-at-onset, and B12 level GWASs, using LD Score Regression with standard settings.[27]

Standard Protocol Approvals, Registrations, and Patient Consents

This study was approved by the Institutional Review Board of McGill University (IRB Study Number A11-M60-21A [21-11-023]). All analyses were conducted using deidentified, previously published, or consortium-controlled data sets. Appropriate institutional and data use agreements were obtained from the respective data access committees.

Data Availability

All supplementary materials, including additional tables and figures, are available in the Supplementary Materials section. The data used in the preparation of this article were obtained from the AMP PD Knowledge Platform[36] and the UKBB via Neurohub.[44] The full GWAS summary statistics for the 23andMe, Inc. discovery data set will be made available through 23andMe to qualified researchers under an agreement with 23andMe that protects the privacy of the 23andMe participants.[43] provides more information and how to access the data. All code used in the analysis is available on our GitHub repository, which can be accessed.[45]

Results

Genetic Analysis Did Not Identify a Role for Common or Rare Variants in Genes Involved in B12 Metabolism in PD Risk

Because the human body cannot synthesize vitamin B12, no genes directly encode for its production. Instead, we focused on key genes involved in B12 metabolism to determine whether dysfunction in these genes may contribute to risk of PD (Table 1; Figure 1). We analyzed common variants using data from the largest available PD risk GWAS[10] and did not find any variants reaching GWAS level of significance (eFigure 1).

We conducted 2 pathway PRS analyses to further evaluate the association between B12 metabolism genes and PD. The first analysis, which focused on 7 key B12 metabolism genes was not significant (odds ratio [OR]) = 1.025 (95% CI 0.999–1.052, $p = 0.061$), while the second, which included all genes reaching GWAS significance for B12 levels, showed

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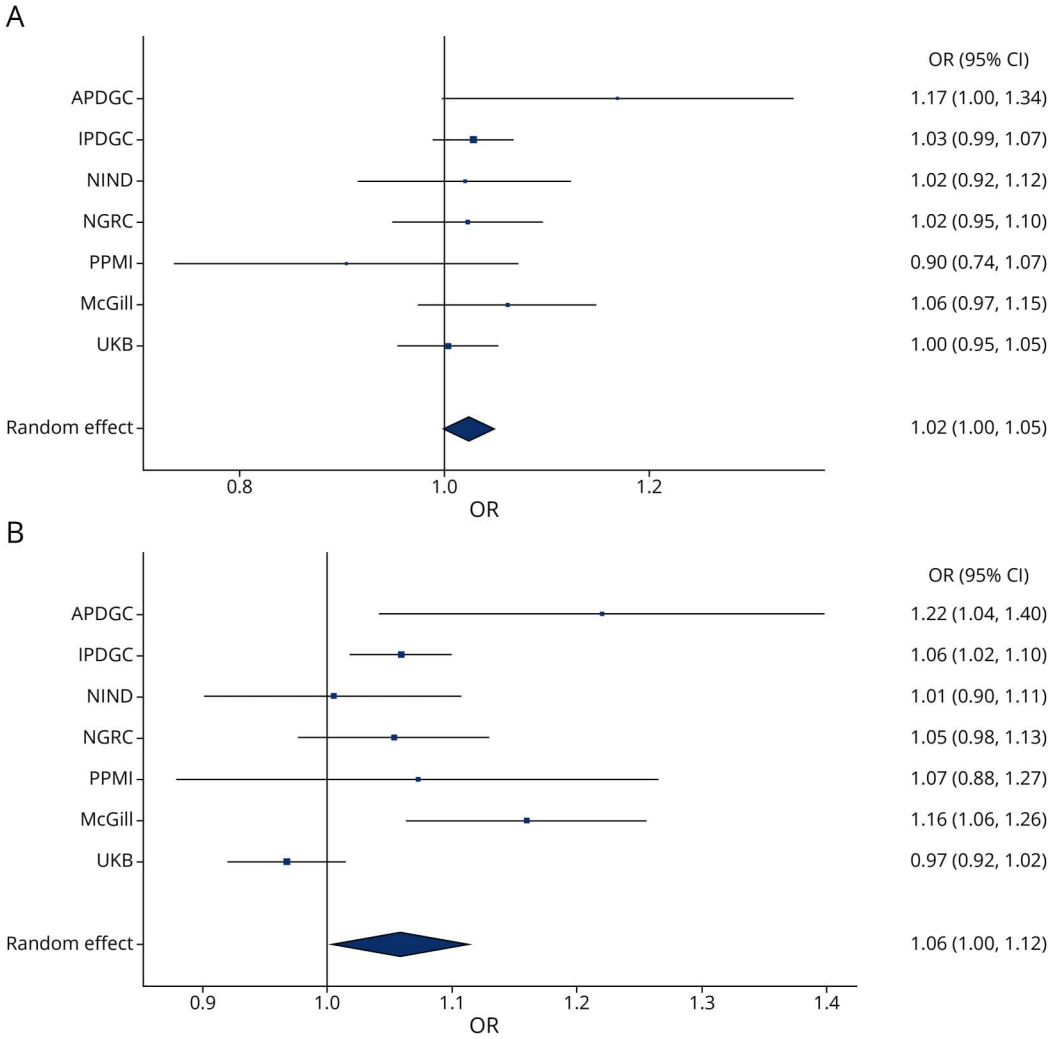
nominal significance (OR = 1.061, 95% CI 1.004–1.121, p = 0.038). After adding *LRRK2* p.G2019S carrier status as a covariate, we did not observe significant differences in our results (seven-gene OR = 1.019, 95% CI 0.965–1.077, p = 0.47; all-loci OR = 1.065, 95% CI 1.014–1.118, p = 0.01), suggesting that the nominal signal is not driven by p.G2019S carriers. No association survived correction for multiple comparisons (Figure 2; eTable 5).

We then analyzed rare variants using SKAT-O in 2 independent cohorts (eTable 6). We identified a significant association between rare variants with high CADD scores in the *CUBN* gene (p = 6.07E-05; Pfdr = 0.005) in the AMP-PD cohort but not in the meta-analysis after correction for multiple comparisons (p = 0.011; Pfdr = 0.292) with PD risk. Notably, the rare variant p.G3114S in *CUBN* was primarily responsible for this association in the AMP-PD cohort (OR = 3.3; p = 3.56E-05); however, this variant was not associated with PD in the UKBB cohort without proxy cases (OR = 1.12; p = 0.22). ClinVar classifies p.G3114S as a benign variant, with a gnomAD MAF of 0.009 in the European population. Given the lack of replication and its relatively high frequency in gnomAD, it is likely that the significance observed in the AMP-PD cohort occurred by chance. In addition, we observed an association between high CADD score variants in *TCN2* (p = 0.006; Pfdr = 0.244) in the UKBB cohort, but not in the AMP-PD cohort or meta-analysis. No other associations were significant.

MR and Genetic Correlation Did Not Reveal Causal or Genetic Link Between B12 Level and PD

The instruments had sufficient power, as indicated by F-statistics >10. The analysis demonstrated sufficient statistical

Figure 2 Two Pathway-Specific PRS Analyses of B12 Metabolism Genes



Panel A shows the pathway-specific PRSs for 7 genes crucial in B12 metabolism. Panel B displays the pathway-specific PRSs for all GWAS-significant genes linked to B12 metabolism. APDGC = Autopsy-Confirmed Parkinson Disease GWAS Consortium; IPDGC = International Parkinson Disease Genomics Consortium; NGRC = NeuroGenetics Research Consortium; NINDSs = NINDSs and Stroke Repository Parkinson's Disease Collection; OR = odds ratio; PPMI = Parkinson's Progression Markers Initiative; UKBB = UK Biobank.

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power (80%) for PD risk and age-at-onset GWASs; however, the parameters related to PD progression had less robust power, falling below the 80% threshold. No potentially causal association between B12 metabolism and PD risk, age at onset, or progression was observed (Table 2). To test for potential violations of MR assumptions, we performed sensitivity analyses (eTable 7). No significant heterogeneity was evident. The results of the Egger intercept test suggest no strong evidence for horizontal pleiotropy. The MR-PRESSO test did not detect any potential pleiotropic outlier SNVs.

We did not find genetic correlation between PD risk, PD age-at-onset, and B12 level GWASs (rg = −0.05, p = 0.69, and rg = 0.12, p = 0.62, respectively; eTable 8).

Discussion

In this study, we comprehensively investigated the genetic role of B12 metabolism and levels in PD. We found no robust evidence to suggest that common or rare variants related to B12 metabolism are associated with PD risk. We also found no potential causal link between B12 levels and PD age at onset or risk of PD, similar to previous MR analysis[7] or PD progression.

Epidemiologic studies suggest that B12 levels tend to be lower in patients with PD compared with healthy controls.[2-4] It is possible that confounding factors, particularly nonmotor symptoms of PD such as gastrointestinal dysfunction, weight loss with disease progression, and decreased oral food intake, could contribute to variations in B12 levels and may obscure any underlying associations. For example, constipation, which affects most patients with PD,[28] along with delayed gastric emptying[29] and decreased oral intake in the later stages of the disease, could interfere with B12 levels. In addition, alterations in gut microbiota in PD[30] may influence prevalence of bacteria responsible for B12 synthesis.[31,32]

Table 2 Results of the MR Analyses Between B12 Metabolism and PD Risk and Progression

			Inverse variance weighted	
Outcome GWAS	**N SNVs**	**Power (%)**	***p* Value**	**OR (95% CI)**
PD	5	100	0.892	1.02 (0.81–1.27)
PD age at onset	2	96.7	0.620	0.54 (0.05–6.27)
CogI	3	21.5	0.181	0.33 (0.06–1.68)
MDS-UPDRS part III	3	18.3	0.331	0.79 (0.49–1.27)
MMSE	3	17.6	0.987	0.99 (0.43–2.3)
HY	3	14.4	0.090	0.87 (0.73–1.02)

Abbreviations: CogI = cognitive impairment; HY = Hoehn and Yahr scale; MDS-UPDRS part III = Unified Parkinson's Disease Rating Scale; MMSE = Mini-Mental State Examination; OR = odds ratio; p = p value; PD = Parkinson disease; R^2 = proportion of variance in exposure variable explained by SNVs.

Vitamin B12 is a central cofactor in one-carbon and mitochondrial metabolism. When B12 is low, homocysteine and methylmalonic acid accumulate, S-adenosyl-methionine declines, and downstream shift in acylcarnitine and phospholipid profiles occurs.[33,34] These changes can drive oxidative stress, DNA hypomethylation, and mitochondrial dysfunction, pathways already implicated in PD. These epigenetic and metabolic intermediates, therefore, represent a plausible nongenetic route by which B12 status could influence PD phenotypes and deserve closer investigation.

While our analyses did not identify a genetic link, it is important to consider certain limitations. First, our analyses were restricted to individuals of European ancestry, which limits the generalizability of our findings to other populations. Second, in the MR analysis, GWAS data on PD progression have been underpowered, reducing our ability to detect associations. Third, for motor progression evaluation in the GWAS outcomes, medication state at examination differed across cohorts. Although the meta-analysis[14] adjusted for dopaminergic therapy, residual ON/OFF heterogeneity may still influence progression estimates. Fourth, we did not evaluate confounding factors that could influence B12 metabolism (e.g., malabsorption syndromes, use of supplements, and anemia). Fifth, although our gene lists were designed to cover each major step of B12 absorption, transport, and activation (Figure 1), the pathway is extensive, and certain genes involved in absorption and intracellular trafficking (e.g., *CBLIF*, *LMBRD1*, and *MMADHC*) were not included. These genes are reasonable candidates for inclusion in future analyses, particularly in larger and multiancestry data sets. We verified that these were not significant loci in the PD GWASs we analyzed.[10] A recent preprint reports multiple novel PD loci, including *LMBRD1*. Because these findings are only recently reported in a preprint, replication and follow-up studies will be needed to examine these signals and their relevance to B12 metabolism.[35] Finally, we were unable to access GWAS summary statistics for *LRRK2*-related PD, limiting our ability to assess the role of B12 in this genetic form. Thus, further studies in diverse populations and with larger data sets are needed to confirm our results.

In summary, our findings suggest that B12 levels or metabolism may not play a prominent genetic role in PD development or progression, yet addressing low B12 levels in patients with PD is important to prevent B12 deficiency–related symptoms. Nongenetic factors, such as gastrointestinal dysfunction and microbiota alterations, may contribute to observed differences in B12 levels in patients with PD. Future research should explore these environmental and physiologic factors more closely to better understand their impact on B12 metabolism in the context of PD.

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Up-to-date information on the study can be obtained from amp-pd.org. AMP-PD—a public-private partnership—is managed by the FNIH and funded by Celgene, GSK, the Michael J. Fox Foundation for Parkinson's Research, the National Institute of Neurological Disorders and Stroke, Pfizer, Sanofi, and Verily. Genetic data used in preparation of this article were obtained from the Fox Investigation for New Discovery of Biomarkers (BioFIND)[37], the Harvard Biomarker Study (HBS)[38], the Parkinson's Progression Markers Initiative (PPMI)[39], the Parkinson's Disease Biomarkers Program (PDBP)[37-40], the International LBD Genomics Consortium (iLBDGC), and the STEADY-PD III Investigators. BioFIND is sponsored by the Michael J. Fox Foundation for Parkinson's Research (MJFF), with support from the National Institute for Neurological Disorders and Stroke (NINDS). The BioFIND Investigators have not participated in reviewing the data analysis or content of the manuscript. Up-to-date information on the study can be obtained from michaeljfox.org/news/biofind. The HBS is a collaboration of HBS investigators [full list of HBS investigators found at bwhparkinsoncenter.org/biobank/and funded through philanthropy and NIH and non-NIH funding sources. The HBS Investigators have not participated in reviewing the data analysis or content of the manuscript. PPMI—a public-private partnership—is funded by the Michael J. Fox Foundation for Parkinson's Research and funding partners (list of the full names of all of the PPMI funding partners found at ppmi-info.org/fundingpartners). The PPMI investigators have not participated in reviewing the data analysis or content of the manuscript. Up-to-date information on the study can be obtained from ppmi-info.org. The PDBP consortium is supported by the NINDS at the NIH. A full list of PDBP investigators can be found at pdbp.ninds.nih.gov/policy. "The PDBP investigators have not participated in reviewing the data analysis or content of the manuscript. Genome Sequencing in Lewy Body Dementia and Neurologically Healthy Controls: A Resource for the Research Community" was generated by the iLBDGC, under the co-directorship by Dr. Bryan J. Traynor and Dr. Sonja W. Scholz from the Intramural Research Program of the U.S. NIH. The iLBDGC Investigators have not participated in reviewing the data analysis or content of the manuscript. A complete list of contributors is given in[41]. STEADY-PD III is a 36-month, Phase 3, parallel-group, placebo-controlled study[42] of the efficacy of isradipine 10 mg daily in 336 participants with early Parkinson disease that was funded by the NINDS and supported by the Michael J Fox Foundation for Parkinson's Research and the Parkinson's Study Group. The STEADY-PD III investigators have not participated in reviewing the data analysis or content of the manuscript. The full list of STEADY PD III investigators can be found at clinicaltrials.gov/ct2/show/NCT02168842. The authors thank the research participants and employees of 23andMe, Inc. for making this work possible. The full GWAS summary statistics for the 23andMe discovery data set will be made available through 23andMe to qualified researchers under an agreement with 23andMe that protects the privacy of the 23andMe participants. For more information and to apply for access to the data, visit research.23andme.com/collaborate/. This research used the NeuroHub[44] infrastructure and was undertaken thanks in part to funding from the Canada First Research Excellence Fund, awarded through the Healthy Brains, Healthy Lives initiative at McGill University, Calcul Québec, and Compute Canada. This research has been conducted using the UK Biobank Resource under Application Number 45551. The UKBB cohort was accessed using Neurohub (mcgill.ca/hbhl/neurohub). Z. Gan-Or is supported by the Fonds de recherche du Québec–Santé (FRQS) Chercheurs-Boursiers award and is a William Dawson Scholar.

Author Contributions

R. Dering: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. M. Onvumere: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. L. Liu: analysis or interpretation of data. P. Huot: study concept or design. Z. Gan-Or: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. K. Senkevich: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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Disclosure

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